



Elias Loomis experiments

The madman who set out to determine the windspeed velocities required to defeather a chicken. <https://digit.in/jan20-21>



Woman in a flying bathtub

The unbelievable story of a woman from Texas who survived a tornado by hiding in a bathtub. <https://digit.in/jan20-22>

scale. The nados are categorised based on how much damage they cause, and while they are associated with nominal wind speeds, the categorisation itself is based on the damage and not the wind speeds. From EFO to EF5, these nados cause light, moderate, considerable, severe, devastating and incredible damage. The description for an EF5 nado essentially says it all, "Strong-framed, well-built houses leveled off foundations and swept away; steel-reinforced concrete structures are critically damaged; tall buildings collapse or have severe structural deformations; cars, trucks, and trains can be thrown approximately 1 mile (1.6 km)." In the UK, the Tornado and Storm Research Organisation (TORRO) scale is used, which ranks tornados from T0 to T11, based on the amount of devastation caused. EF5 corresponds with a T10+ tornado. Over a 100 EF5 nados have been recorded since 1900.

THE FREAKY EVENTS

In 1764, one of the most devastating tornados in recorded history hit Woldgeck in Germany. The path of the tornado was traced by Gottlob Burchard Genzmer for 30 kilometers. It was a landscape of pure devastation. On the way, Genzmer interviewed the survivors and compiled a 56 page report of his observations, which provide one of the most interesting stories of an early EF5 tornado. Flocks of geese fell from the sky after being struck with hailstones, children straight up disappeared never to be found again, walls of houses got torn out, and some people simply disintegrated. The most telling detail on the intensity of the storm was that chopped down trees, with just the stumps sticking out of the ground, were ripped out of the ground by the passing tornado. While empirically reporting what he found at the site, Genzmer could not understand what had happened and does not even attempt to offer an explanation.

In 1800, an EF5 storm hit Hainichen in Germany, and by then the Germans had come up with a name for the phenomenon. Google translate

renders these names as wind trousers or land trombones. Houses, cows and trees were thrown out of the ground, when not pulverised. The cattle are described as roaring in the air. The tornado lasted only a hour, but made a trip through a forest between two settlements. During its time in the forest, the tornado uprooted trees and hurled them into a nearby river. Some trees had their bark stripped off.

In these earliest storms, the most affected were the livestock. One of the bizarre occurrences after the passing of the tornado, was that chickens would be plucked of all feathers. It became somewhat of a rural myth that the low pressure areas at the hearts of tornadoes were responsible for the defeathering of chicken. In 1842, a scientist by the name of Elias Loomis conducted one of the craziest scientific experiments ever. He fired live



Credit: The tornado project

A rooster allegedly blown into a jug by the 1915 tornado at Kansas

chicken from a cannon, to estimate the required wind speeds to defeather it. At 550 kilometers per hour, the chicken was defeathered, but was also pulverised. Loomis concluded that a more reasonable 160 kilometer per hour wind was required to defeather a chicken, and keep it alive as well, but never came around to actually testing this theory, fortunately for the chicken. He also placed chicken in vacuum jars to see if their feathers would explode off them, and they did not. Scientists now believe that the stressed chicken voluntarily shed the feathers as a response known as moult flight. The behavior would leave a predator with a mouthful of feathers and no chicken.

In 1915, a tornado swept through Kansas where four horses tied up to a rail were lifted off, and transported by a distance of 400 meters. All the horses were found alive, and still tied to the rail. The tornado seemed to have a mind of its own, devastating stone walls but leaving wooden structures untouched, or clearing away houses, but carefully avoiding shelves with neatly stacked items that survived untouched. For over a 100 kilometers, the tornado rained all kinds of human made debris, including photographs, clothes, receipts, money and books. Apparently a rooster was blown into a jug, with only its head sticking out.

The vortices of tornados suck up material, and these then subsequently rain down around its path. The tornado basically works like a vacuum cleaner. Scientists have seen entire ponds being taken into the air by tornados, and it is reasonable to assume that the creatures living within are the cause of reports of animal rain. In 1873, it rained frogs in Kansas, and scientists now believe that this was caused by a tornado. In 1882, it rained frozen frogs in Iowa, which can be considered frog hail. However, no one has actually recorded frogs being sucked by tornados.

In 1947, there were reports of rains of fish in Louisiana. There have also been records of rains of rats.

There are some other crazy reports as well. Cornstalks have been blown into truck radiators, or plastic straws blown into metal sheetings. Cornstalks have been enclosed in ice in the air, subsequently raining down in the form of cornhail. In one tornado, a half liter glass bottle was blown into a one liter bottle without any cracks. In 1976, a house was turned on its side, but remained intact. In 2017, a woman hid from a tornado in the bathtub of her house. Her house was destroyed, and the bathtub deposited in the nearby woods. That must have been one interesting flight. 📌